

1	What are the major products of the pentose phosphate pathway?
1	ATP and NADH
2	NADH and FADH ₂
✓ 3	NADPH and ribose-5-phosphate
4	Pyruvate and acetyl-CoA

2	What is the clinical consequence of excess lactate production during exercise?
1	Hyperglycemia
2	Metabolic alkalosis
✓ 3	Muscle fatigue
4	Hypercalcemia

3	A medical student monitors a diabetic patient's glucose levels. Insulin promotes glucose uptake into adipose tissue via which transporter?
1	GLUT-1
2	GLUT-2
3	GLUT-3
✓ 4	GLUT-4

4	A 3-year-old child with fatigue and hemolytic anemia is found to have reduced ATP levels in red cells. The most likely enzyme deficiency is:
1	Phosphoenolpyruvate carboxykinase
✓ 2	Pyruvate kinase
3	Aldolase
4	Enolase

5	What is the location of the pyruvate dehydrogenase (PDH) complex?
1	Cytoplasm
2	Endoplasmic reticulum
✓ 3	Mitochondrial matrix
4	Intermembrane space

INTRODUCTION

4 Intermembrane space

6 What is the immediate precursor for glycogen synthesis?

1 Glucose-6-phosphate

2 UDP-glucose

3 Glucose-1-phosphate

4 ATP

7 Which enzyme is deficient in Pompe disease (Type II glycogen storage disease)?

1 Glycogen synthase

2 Acid maltase

3 Branching enzyme

4 Debranching enzyme

8 Glucose-6-phosphate dehydrogenase (G6PD) deficiency leads to:

1 Hyperglycemia

2 Hemolytic anemia

3 Hyperlipidemia

4 Ketoacidosis

9	Classic galactosemia is caused by a deficiency of:
✓ 1	Galactokinase
2	Galactose-1-phosphate uridylyltransferase
3	UDP-glucose pyrophosphorylase
4	Hexokinase

10	In patients with liver failure, gluconeogenesis is impaired. What is a likely consequence?
✓ 1	Persistent hyperglycemia
2	Recurrent hypoglycemia
3	Increased muscle mass
4	Increased fat storage

11	Which metabolic pathway is upregulated in response to increased ADP and Ca^{2+} during exercise?
✓ 1	Glycogenolysis
2	Citric acid cycle
3	Fatty acid synthesis
4	Urea cycle

12	Which condition promotes activation of pyruvate dehydrogenase (PDH)?
1	High ATP
2	High NADH
✓ 3	Insulin
4	Acetyl-CoA

13	Maple syrup urine disease is due to a defect in the degradation of which amino acid?
1	Tyrosine
2	Methionine
3	Tryptophan
✓ 4	Isoleucine

4 Isoleucine

14	Which statement about transamination reactions is correct?
1	They involve ATP hydrolysis
2	They are irreversible
3	They require NAD ⁺ or NADP ⁺
✓ 4	They require pyridoxal-5'-phosphate (PLP)

15	Which amino acids are considered purely ketogenic?
✓ 1	Lysine and leucine
2	Tryptophan and phenylalanine
3	Histidine and lysine
4	Isoleucine and leucine

16	Which statement about the urea cycle is correct?
1	Argininosuccinate is cleaved to urea and ornithine
2	Carbamoyl phosphate provides both nitrogens of urea
✓ 3	Arginine is hydrolyzed to urea and ornithine

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رشته تخصصی:

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دانشکده:

4	The urea cycle generates energy
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4 The urea cycle generates energy

17 Histamine is synthesized from histidine by which process?

- 1 Decarboxylation
- 2 Demethylation
- 3 Hydrogenation
- 4 Oxygenation

18 Amino acids are absorbed into the portal circulation by:

- 1 Pinocytosis
- 2 Active transport
- 3 Facilitated diffusion
- 4 Passive diffusion

19 In the ATP-dependent proteasome system, which protein is required to mark proteins for degradation?

- 1 Cyclin
- 2 Cytochrome
- 3 Ubiquitin
- 4 Histone

20 A 52-year-old man has dark urine, and homogentisic acid is detected in 24-hour urine. What is the diagnosis?

- 1 Albinism
- 2 Pheochromocytoma
- 3 Hartnup disease
- 4 Alkaptonuria

21 Which plasma enzyme is normally found in urine?

- 1 Acid phosphatase

21	Which plasma enzyme is normally found in urine?	
	1	Acid phosphatase
	2	Nucleotidase
	3	Alkaline phosphatase
✓	4	Amylase

22	The most important cause of increased serum aminotransferase activity is:	
	1	Bone disease
✓	2	Liver disease
	3	Pancreatitis
	4	Brain disease

23	Which of the following is NOT a pancreatic-specific enzyme?	
	1	Lipase
✓	2	Aldolase
	3	Trypsin
	4	Amylase

24	Which enzyme is measured to diagnose rhabdomyolysis?	
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24

Which enzyme is measured to diagnose rhabdomyolysis?

صفحه ۴

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دانشکده:

	1	Lipase
	2	Nucleotidase
✓	3	Creatine kinase
	4	Gamma-glutamyltransferase

25

Which statement about high-density lipoprotein (HDL) is correct?

	1	HDL is synthesized only in the liver
	2	HDL transports triacylglycerols to peripheral tissues
✓	3	HDL picks up cholesterol and converts it to cholesteryl esters
	4	HDL cannot be endocytosed by the liver

26

The key regulatory enzyme of cholesterol synthesis is:

	1	HMG-CoA synthase
	2	HMG-CoA lyase
✓	3	HMG-CoA reductase
	4	Mevalonate kinase

27	All are conditions associated with ketosis EXCEPT :	
	1	Starvation
	2	Uncontrolled diabetes mellitus
✓	3	High-carbohydrate diet
	4	Alcohol consumption

28	All are ketone bodies EXCEPT :	
	1	Acetone
	2	Acetoacetate
	3	Beta-hydroxybutyrate
✓	4	Lactic acid

29	Which anti-obesity drug inhibits gastric and pancreatic lipases, decreasing fat absorption?	
	1	Ezetimibe
✓	2	Orlistat
	3	Statin
	4	Bile acid resin

30	Pancreatic juice does NOT contain:	
✓	1	Bile salts
	2	Phospholipase A ₂
	3	Lipase and colipase
	4	Lipoprotein lipase

31	Which drug reduces cholesterol synthesis by inhibiting HMG-CoA reductase?	
	1	Orlistat
✓	2	Statins
	3	Ezetimibe
	4	Choline

32	Which statement about β -oxidation of fatty acids is TRUE?
1	It occurs in the cytoplasm
2	It is stimulated by insulin
✓ 3	It occurs in mitochondria and generates acetyl-CoA
4	It leads directly to glucose synthesis
33	What is the main function of lipoprotein lipase (LPL)?
1	Synthesizes triglycerides in the liver
✓ 2	Hydrolyzes triglycerides in chylomicrons and VLDL
3	Transports cholesterol into hepatocytes
4	Converts cholesterol to bile acids
34	During prolonged fasting, the brain primarily uses:
1	Glucose
2	Fatty acids
✓ 3	Ketone bodies
4	Glycogen
35	Which condition increases plasma VLDL levels?
1	High-fiber diet
✓ 2	High-carbohydrate intake
3	Fasting
4	Aerobic exercise
36	What is true about short- and medium-chain fatty acids?
1	They require micelles for absorption
✓ 2	They are absorbed directly into portal blood
3	They are poorly absorbed
4	They form lipoproteins for absorption

	2	They are absorbed directly into portal blood
	3	They are poorly absorbed
	4	They form lipoproteins for absorption

37	Deficiency of which enzyme causes immunodeficiency in newborns?	
	1	Thymidylate synthase
✓	2	Adenosine deaminase
	3	Xanthine oxidase
	4	Ribonucleotide reductase

38	All are involved in pyrimidine synthesis EXCEPT :	
	1	Aspartate
	2	Phosphoribosyl pyrophosphate
	3	Carbamoyl phosphate
✓	4	Glycine

39	5-Fluorouracil inhibits:	
	1	Ribonucleotide reductase
	2	Dihydrofolate reductase
✓	3	Thymidylate synthase
	4	Xanthine oxidase

40	Lesch-Nyhan syndrome is caused by deficiency of
✓ 1	Hypoxanthine-guanine phosphoribosyltransferase
2	Adenosine deaminase
3	Xanthine oxidase
4	Adenine phosphoribosyltransferase

41	Porphyria cutanea tarda is due to deficiency of:
1	ALA synthase
2	Ferrochelatase
✓ 3	Uroporphyrinogen decarboxylase
4	Coproporphyrinogen oxidase

42	In which condition does urinary urobilinogen increase?
1	Dubin-Johnson syndrome
2	Biliary obstruction
3	Hepatitis
✓ 4	Hemolytic anemia

43	Which enzyme opens the porphyrin ring during heme degradation?
✓ 1	Heme oxygenase
2	Biliverdin reductase
3	UDP-glucuronosyltransferase
4	Ferrochelatase

44	The end product of urobilinogen metabolism in the kidney is:
1	Urobilinogen
✓ 2	Urobilin
3	Stercobilin
4	Bilirubin

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1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

3	Stercobilin
4	Bilirubin

45 In the fed state, which is correct?

1	Decreased fatty acid synthase activity
2	Decreased acetyl-CoA carboxylase activity
✓ 3	Decreased carnitine palmitoyltransferase I activity
4	Decreased HMG-CoA reductase activity

46 Which enzyme hydrolyzes stored triacylglycerols in adipose tissue?

1	Acetyl-CoA carboxylase
2	Fatty acid synthase
✓ 3	Hormone-sensitive lipase
4	Lipoprotein lipase

47 The primary role of the liver during fasting is:

1	Glycogen synthesis
✓ 2	Gluconeogenesis
3	Fatty acid oxidation
4	Ketone body synthesis

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48	During fasting, the release of fatty acids from adipose tissue is promoted by:	
	1	Increased insulin and decreased epinephrine
	2	Re-esterification of fatty acids into triglycerides
	3	Activation of lipoprotein lipase
	4	Decreased insulin and increased epinephrine

49	The ATP/ADP translocator in the inner mitochondrial membrane is inhibited by:	
	1	Oligomycin
	2	Cyanide
	3	Atractyloside
	4	Barbiturates

50	Which respiratory chain complex does NOT function as a proton pump?	
	1	Succinate-Q reductase (Complex II)
	2	NADH dehydrogenase (Complex I)
	3	Cytochrome c oxidase (Complex IV)
	4	Cytochrome c reductase (Complex III)

51	Where are FAD and cytochrome aa ₃ located in the electron transport chain?	
	1	Complex I and III
	2	Complex II and IV
	3	Complex II and III
	4	Complex III and IV

52	In all types of oxidoreductases, oxygen is involved <u>EXCEPT</u> in:	
	1	Oxidases
	2	Hydroperoxidases
	3	Dehydrogenases
	4	Oxygenases